

40 Ejercicios del Libro “Operational Research” Wayne L.Winston

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EJERCICIO 1

61[†] E.J. Korvair Department Store has \$1,000 in available cash. At the beginning of each of the next six months, E.J. will receive revenues and pay bills as shown in Table 85. It is clear that E.J. will have a short-term cash flow problem until the store receives revenues from the Christmas shopping season. To solve this problem, E.J. must borrow money.

At the beginning of July, E.J. may take out a six-month loan. Any money borrowed for a six-month period must be paid back at the end of December along with 9% interest (early payback does not reduce the interest cost of the loan). E.J. may also meet cash needs through month-to-month borrowing. Any money borrowed for a one-month period incurs an interest cost of 4% per month. Use linear programming

TABLE 85

Month	Revenues (\$)	Bills (\$)
July	1,000	5,000
August	2,000	5,000
September	2,000	6,000
October	4,000	2,000
November	7,000	2,000
December	9,000	1,000

to determine how E.J. can minimize the cost of paying its bills on time.

```
#conjuntos
set MESES ordered;

#parametros
param ingresos{MESES};
param gastos{MESES};
param efectivo_inicial;
param tasa_corto_plazo default 0.04;
param tasa_largo_plazo default 0.09;

param anterior{MESES} symbolic;

#variables
var corto{MESES} >= 0;
var largo >= 0;
var efectivo{MESES} >= 0;

subject to balance_julio:
    efectivo["Julio"] = efectivo_inicial + ingresos["Julio"] + corto["Julio"] + largo - gastos["Julio"];

subject to balanceresto{m in MESES: m != "Julio":
    efectivo[m] = efectivo[anterior[m]] + ingresos[m] + corto[m] - gastos[m];

subject to pago_largo:
    efectivo["Diciembre"] >= largo * (1 + tasa_largo_plazo);

minimize costo_total:
    sum{m in MESES} corto[m]*tasa_corto_plazo + largo*tasa_largo_plazo;
```

```

ampl: model problema1.mod;
ampl: data problema1.dat;
ampl: option solver cplex;
ampl: solve;
CPLEX 22.1.2:          CPLEX 22.1.2: optimal solution; objective 400
0 simplex iterations
ampl: display costo_total;
costo_total = 400

ampl: display pago_largo;
pago_largo = 0

ampl: display largo;
largo = 0

ampl: display efectivo;
efectivo [*] :=
    Julio  7000
    Agosto 4000
Septiembre  0
    Octubre 2000
    Noviembre 7000
    Diciembre 15000
;

ampl: display corto;
corto [*] :=
    Julio 10000
    Agosto  0
Septiembre  0
    Octubre  0
    Noviembre  0
    Diciembre  0
;

```

E.J. puede cubrir todos sus gastos mensuales, manteniendo saldos positivos en cada mes, y minimiza el costo total de financiamiento a \$400. Esto demuestra una planificación eficiente que garantiza liquidez mensual y el menor costo posible.

EJERCICIO 2

62⁺ Olé Oil produces three products: heating oil, gasoline, and jet fuel. The average octane levels must be at least 4.5 for heating oil, 8.5 for gas, and 7.0 for jet fuel. To produce these products Olé purchases two types of oil: crude 1 (at \$12 per barrel) and crude 2 (at \$10 per barrel). Each day, at most 10,000 barrels of each type of oil can be purchased.

Before crude can be used to produce products for sale, it must be distilled. Each day, at most 15,000 barrels of oil can be distilled. It costs 10¢ to distill a barrel of oil. The result of distillation is as follows: (1) Each barrel of crude 1 yields 0.6 barrel of naphtha, 0.3 barrel of distilled 1, and 0.1 barrel of distilled 2. (2) Each barrel of crude 2 yields 0.4 barrel of naphtha, 0.2 barrel of distilled 1, and 0.4 barrel of distilled 2. Distilled naphtha can be used only to produce gasoline or jet fuel. Distilled oil can be used to produce heating oil or it can be sent through the catalytic cracker (at a cost of 15¢ per barrel). Each day, at most 5,000 barrels of distilled oil can be sent through the cracker. Each barrel of distilled 1 sent through the cracker yields 0.8 barrel of cracked 1 and 0.2 barrel of cracked 2. Each barrel of distilled 2 sent through the cracker yields 0.7 barrel of cracked 1 and 0.3 barrel of cracked 2. Cracked oil can be used to produce gasoline and jet fuel but not to produce heating oil.

The octane level of each type of oil is as follows: naphtha, 8; distilled 1, 4; distilled 2, 5; cracked 1, 9; cracked 2, 6.

All heating oil produced can be sold at \$14 per barrel; all gasoline produced, \$18 per barrel; and all jet fuel produced, \$16 per barrel. Marketing considerations dictate that at least 3,000 barrels of each product must be produced daily. Formulate an LP to maximize Olé's daily profit.

```
maximize beneficio:
    6.5 * aceite + 8.5 * gasolina + 7.0 * avion
    - sum{i in CRUDO} costo_crudo[i] * x[i]
    - costo_destilacion * sum{i in CRUDO} d[i]
    - costo_desintegra * aceite_procesado;

subject to limite_crudo{i in CRUDO}:
    x[i] <= max_diario[i];

subject to limite_destilacion:
    sum{i in CRUDO} d[i] <= max_destilar;

subject to crudo_usado{i in CRUDO}:
    d[i] <= x[i];

subject to calculo_dest1:
    dest1_total = sum{i in CRUDO} rendimiento_dest1[i] * d[i];

subject to limite_desintegra:
    aceite_procesado <= max_desintegra;

subject to origen_aceite:
    aceite_procesado <= dest1_total;

subject to salida_frac1:
    frac1 = rendimiento_frac1 * aceite_procesado;

subject to salida_frac2:
    frac2 = rendimiento_frac2 * aceite_procesado;

# Productos
subject to calculo_aceite:
    aceite = (1 - fraccion_dest1_usable) * dest1_total;

subject to calculo_gasolina:
    gasolina = fraccion_dest1_usable * dest1_total + rendimiento_nafta_desintegra * aceite_procesado;

subject to calculo_avion:
    avion = frac1 + frac2;
```

```
set CRUDO; # tipos de crudo: 1 y 2

param costo_crudo{CRUDO};
param max_diario{CRUDO};

param rendimiento_nafta{CRUDO};
param rendimiento_dest1{CRUDO};
param rendimiento_dest2{CRUDO};

param max_destilar;
param costo_destilacion;

param max_desintegra;
param costo_desintegra;

param fraccion_dest1_usable;
param rendimiento_nafta_desintegra;
param rendimiento_frac1;
param rendimiento_frac2;

param indice_min_aceite;
param indice_min_gasolina;
param indice_min_avion;

param indice_octano;

# Variables de decisión
var x{CRUDO} >= 0;
var d{CRUDO} >= 0;

var dest1_total >= 0;
var aceite_procesado >= 0;
var frac1 >= 0;
var frac2 >= 0;

# Productos finales
var aceite >= 0;
var gasolina >= 0;
var avion >= 0;
```

```

ampl: model problema2.mod;
ampl: data problema2.dat;
ampl: option solver cplex;
ampl: solve;
CPLEX 22.1.2:          CPLEX 22.1.2: optimal solution; objective 0
0 simplex iterations
ampl: display aceite;
aceite = 0

ampl: display gasolina;
gasolina = 0

ampl: display avion;
avion = 0

ampl: display beneficio;
beneficio = 0

ampl: display aceite_procesado;
aceite_procesado = 0

```

En este caso, el modelo indica que lo mejor para la empresa es no hacer nada: no comprar crudo, no procesarlo y no producir ningún producto. debido a que los costos de operación son más altos que las ganancias por ende el modelo concluye que es mejor no hacer nada

EJERCICIO 3

63 Donald Rump is the international funds manager for Countribank. Each day Donald's job is to determine how the bank's current holdings of dollars, pounds, marks, and yen should be adjusted to meet the day's currency needs. Today the exchange rates between the various currencies are given in Table 86. For example, one dollar can be converted to .58928 pounds, or one pound can be converted to 1.697 dollars.

At the beginning of the day, Countribank has the currency holdings given in Table 87.

At the end of the day, Countribank must have at least the amounts of each currency given in Table 88.

Donald's goal is to each day transfer funds in a way that makes currency holdings satisfy the previously listed mini-

TABLE 86

From	To			
	Dollars	Pounds	Marks	Yen
Dollars	1	.58928	1.743	138.3
Pounds	1.697	1	2.9579	234.7
Marks	.57372	.33808	1	79.346
Yen	.007233	.00426	.0126	1

²Based on Garvin et al. (1957).

TABLE 87

Currency	Units (in Billions)
Dollars	8
Pounds	1
Marks	8
Yen	0

TABLE 88

Currency	Units (in Billions)
Dollars	6
Pounds	3
Marks	1
Yen	10

mums, and maximizes the dollar value of the currency holdings at the end of the day.

To figure out the dollar value of, say, one pound, average the two conversion rates. Thus, one pound is worth approximately

$$\frac{1.697 + (1/.58928)}{2} = 1.696993 \text{ dollars}$$

```

set MONEDAS;

param tasa_cambio{MONEDAS};
param valor_inicial{MONEDAS};
param minimo_requerido{MONEDAS};

var x{MONEDAS, MONEDAS} >= 0;

var saldo_final{MONEDAS};

var valor_dolares;

#Conservación de saldos
s.t. Conservacion{j in MONEDAS}:
    saldo_final[j] = sum{i in MONEDAS} x[i,j] - sum{k in MONEDAS} x[j,k] + valor_inicial[j];

#minimos requeridos
s.t. Minimo{j in MONEDAS}:
    saldo_final[j] >= minimo_requerido[j];

#no transferir más de lo disponible
s.t. LimiteTransferencias{i in MONEDAS}:
    sum{j in MONEDAS} x[i,j] <= valor_inicial[i];

# Cálculo del valor total en dólares
s.t. Objetivo:
    valor_dolares = sum{j in MONEDAS} tasa_cambio[j] * saldo_final[j];

maximize MaxValor: valor_dolares;

```

```

ampl: reset;
model problema3.mod;
data problema3.dat;
option solver cplex;
solve;
display x, saldo_final, valor_dolares;
CPLEX 22.1.2:          CPLEX 22.1.2: optimal solution; objective 3301840
5 simplex iterations
x :=
dolar dolar      0
dolar euro       500
dolar peso       0
dolar real       0
euro dolar       0
euro euro        0
euro peso        0
euro real        0
peso dolar       0
peso euro        3e+06
peso peso        0
peso real        0
real dolar       300
real euro        700
real peso        0
real real        0
;

saldo_final [*] :=
dolar      300
euro 3001400
peso       0
real       0
;

valor_dolares = 3301840

```

Se cumplió con los mínimos requeridos de cada moneda y se alcanzó un valor total óptimo de 3.301.840 dólares, mostrando una gestión eficiente y rentable del capital..

EJERCICIO 4

16 Suppose my utility function for my asset position is $u(x) = x^{1/2}$. I have \$10,000 at present. Consider the following lottery:

L : With probability $\frac{1}{2}$, L yields a payoff of \$1,025.

With probability $\frac{1}{2}$, L yields a payoff of $-\$199$.

a If I don't have the right to play L , find an equation that when solved would yield the amount I would be willing to pay for the right to play L . This is called the **buying price** of lottery L .

b If I have the right to play L , what is the least I would accept from somebody who wanted to buy the right to play L ? (After someone else buys L , I can't play L .) This is called the **selling price** of lottery L .

c Answer part (b) for the case that I have \$1,000.

d Suppose that my utility function for my asset position is $u(x) = 1 - e^{-x}$. Show that for all possible asset positions, the buying price of L and the selling price of L will remain the same. Show that for all asset positions, the buying price of L will equal the selling price of L .

```
param ahora;      # posición actual
param gana;
param pierde;

var p >= 0;        # precio de compra
var q >= 0;        # precio de venta

subject to buying_price:
    1 - exp(-(ahora - p)) =
        0.5 * (1 - exp(-(ahora + gana))) +
        0.5 * (1 - exp(-(ahora + pierde)));

subject to selling_price:
    1 - exp(-ahora) =
        0.5 * (1 - exp(-(ahora + gana - q))) +
        0.5 * (1 - exp(-(ahora + pierde - q)));
```

EXIT: Optimal Solution Found.

```
model problema4.mod;
data problema4.dat;
option solver ipopt;
solve;
display p;

Ipopt 3.12.13: Optimal Solution Found
suffix ipopt_zU_out OUT;
suffix ipopt_zL_out OUT;
p = 0.00999999
```

El modelo calculó que esta persona solo pagaría alrededor de un centavo por el derecho a jugar. En otras palabras, prefiere no arriesgar su dinero, y esa pequeña cantidad refleja su actitud cautelosa ante la incertidumbre.

EJERCICIO 5

8 State University is about to play Ivy College for the state tennis championship. The State team has two players (A and B), and the Ivy team has three players (X, Y, and Z). The following facts are known about the players' relative abilities:

X will always beat B; Y will always beat A; A will always beat Z. In any other match, each player has a $\frac{1}{2}$ chance of winning. Before State plays Ivy, the State coach must determine who will play first singles and who will play second singles. The Ivy coach (after choosing which two players will play singles) must also determine who will play first singles and second singles. Assume that each coach wants to maximize the expected number of singles matches won by the team. Use game theory to determine optimal strategies for each coach and the value of the game to each team.

```
set State_es;      #estrategias de State
set Ivy_es;        #estrategias de Ivy

param posibilidad{State_es, Ivy_es};

var p {i in State_es} >= 0;
var v;

s.t. prob_sum: sum {i in State_es} p[i] = 1;

s.t. min_valor {j in Ivy_es}:
    sum {i in State_es} p[i] * posibilidad[i,j] >= v;

maximize valor_esp: v;

ampl: reset;
model problema5.mod;
data problema5.dat;
option solver cplex;
solve;
display p, v;
CPLEX 22.1.2:                CPLEX 22.1.2: optimal solution; objective 0.5
0 simplex iterations
p [*] :=
AB  0
BA  1
;

v = 0.5
```

El modelo encuentra que la mejor estrategia para State es jugar siempre con su jugador A en ambos partidos individuales. Esta táctica les da, en promedio, una victoria por encuentro es decir, ganan la mitad de los partidos

EJERCICIO 6

11 A drug store sells 30 bottles of antibiotics per week. Each time it orders antibiotics, there is a fixed ordering cost of \$10 and a cost of \$10/bottle. Assume that the annual holding cost is 20% of the cost of a bottle of antibiotics, and suppose antibiotics spoil and cannot be sold if they spend more than one week in inventory. When the drug store places an order, how many bottles of antibiotics should be ordered?

```
param D;      # demanda semanal
param S;      # costo fijo por pedido
param C;      # costo unitario
param h;      # costo semanal de inventario por unidad

var q >= 1, <= D; # cantidad de botellas pedidas

# Costo total por semana como función del valor de q
minimize costo_total:
    (D / q) * S + D * C + (q / 2) * h;

ampl: reset;
model problema6.mod;
data problema6.dat;
option solver ipopt;
solve;
display q, costo_total;
Ipopt 3.12.13:

EXIT: Optimal Solution Found.

Ipopt 3.12.13: Optimal Solution Found

suffix ipopt_zU_out OUT;
suffix ipopt_zL_out OUT;
q = 30
costo_total = 310.577
```

El modelo muestra que lo mejor es pedir exactamente 30 botellas por semana, justo lo que se vende con un gasto total semanal de aproximadamente \$310.58

EJERCICIO 7

16 The borough of Staten Island has two “sanitation districts.” In district 1, street litter piles up at an average rate of 2,000 tons per week, and in district 2 at an average rate of 1,000 tons per week. Each district has 500 miles of streets. Staten Island has 10 sanitation crews and each crew can clean 50 miles per week of streets. To minimize the average level of the total amount of street litter in the two districts, how often should each district be cleaned? Assume that litter in a district grows at a constant rate until it is picked up (assume pickup is instantaneous). (*Hint: Let p_i equal the average number of times that each district is cleaned per week. Then $p_1 + p_2 = 1$.*)[†]

```
set D; #distritos

param tasa_acumulacion {D};
param longitud_calles {D};
param limpieza_total;

var p {D} >= 0; #frecuencia de limpieza

s.t. suma_frecuencias: sum {i in D} p[i] = 1;

s.t. limite_cuadrillas: sum {i in D} p[i] * longitud_calles[i] <= limpieza_total;

minimize basura_total:
    sum {i in D} tasa_acumulacion[i] / (2 * p[i]);

ampl: model problema7.mod
ampl: data problema7.dat
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 2914.213562
8 simplex iterations
1 branching node
absmipgap=0.225849, relmipgap=7.74992e-05
ampl: display basura_total;
basura_total = 2914.21
```

se logra reducir la basura acumulada a unas 2.91 toneladas en promedio, aprovechando al máximo los recursos disponibles

EJERCICIO 8

4 A hospital orders its thermometers from a hospital supply firm. The cost per thermometer depends on the order size q , as shown in Table 6. The annual holding cost is 25% of the purchasing cost. Let EOQ_{80} be the EOQ if the cost per thermometer is 80¢, and let EOQ_{79} be the EOQ if the cost per thermometer is 79¢.

- Explain why EOQ_{79} will be larger than EOQ_{80} .
- Explain why the optimal order quantity must be either EOQ_{79} , EOQ_{80} , or 100.
- If $EOQ_{80} > 100$, show that the optimal order quantity must be EOQ_{79} .
- If $EOQ_{80} < 100$ and $EOQ_{79} < 100$, show that the optimal order quantity must be either EOQ_{80} or 100.
- If $EOQ_{80} < 100$ and $EOQ_{79} > 100$, show that the optimal order quantity must be EOQ_{79} .

5 In Problem 4, suppose the cost per order is \$1 and the monthly demand is 50 thermometers. What is the optimal order quantity? How small a discount could the supplier offer and still have the hospital accept the discount?

TABLE 6

Order Size	Price per Thermometer
$q < 100$	80¢
$q \geq 100$	79¢

```

param D;      #demanda anual
param K;      #costo
param c1;     #precio sin descuento
param c2;     #precio con descuento

param qmin := 1;
param qmax := 10000;

var q >= qmin, <= qmax;
var y binary;

var c;
var h;

s.t. activar_descuento: q >= 100 - (1 - y) * qmax;
s.t. limitar_descuento: q <= 99 + y * qmax;

s.t. definir_c: c = c1 * (1 - y) + c2 * y;
s.t. definir_h: h = 0.25 * c;

minimize TotalCost:
    K * D / q + h * q / 2 + c * D;

ampl: model problema8.mod;
ampl: data problema8.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 4394.425282
10 simplex iterations
3 branching nodes
absmipgap=0.306649, relmipgap=6.97813e-05
ampl: display TotalCost;
TotalCost = 4394.43

```

El resultado muestra que conviene pedir al menos 100 termómetros por orden así, el hospital logra un costo total óptimo de aproximadamente \$4394.43 al año.

EJERCICIO 9

4 A company has the option of purchasing a good or manufacturing the item. If the item is purchased, the company will be charged \$25 per unit plus a cost of \$4 per order. If the company manufactures the item, it has a production capacity of 8,000 units per year. It costs \$50 to set up a production run, and annual demand is 3,000 units per year. If the annual holding cost is 10% and the cost of manufacturing one unit is \$23, determine whether the company should purchase or manufacture the item.

```
param D;      # demanda anual (unidades)
param Cp;     # costo unitario si se compra
param Co;     # costo por orden de compra
param Cf;     # costo fijo por producción (setup)
param Cm;     # costo unitario de fabricar
param h;      # tasa de almacenamiento anual
param P;      # tasa de producción (unidades por año)

var metodo binary; # 0 = comprar, 1 = fabricar
var q >= 1;        # tamaño de lote

# Costos en cada caso
param Hc := h * Cp;                # costo de inventario si compra
param Hm := h * Cm * (1 - D/P);    # costo de inventario si fabrica (EPQ)

# Costos totales
var Ccompra;      # costo total si compra
var Cfabricar;    # costo total si fabrica

s.t. calc_compra:
    Ccompra = (D / q) * Co + D * Cp + (q / 2) * Hc;

s.t. calc_fabricar:
    Cfabricar = (D / q) * Cf + D * Cm + (q / 2) * Hm;

var costo_total;

s.t. seleccion:
    costo_total = (1 - metodo) * Ccompra + metodo * Cfabricar;

minimize objetivo: costo_total;

model problema9.mod;
data problema9.dat;
option solver ipopt; # mejor para estabilidad numérica
solve;
display metodo, q, costo_total;
Ipopt 3.12.13;
```

EXIT: Optimal Solution Found.

Ipopt 3.12.13: Optimal Solution Found

```
suffix ipopt_zU_out OUT;
suffix ipopt_zL_out OUT;
metodo = 1
q = 456.832
costo_total = 69656.7
```

El modelo sugiere producir en lotes de aproximadamente 457 unidades, lo que lleva a un costo total cercano a \$69,657

EJERCICIO 10

3 Suppose that instead of measuring shortage in terms of cost per shortage year, a cost of S dollars is incurred for each unit the firm is short. This cost does not depend on the length of time before the backlogged demand is satisfied. Determine a new expression for $TC(q, M)$, and explain how to determine optimal values q^* and M^* .

```
param D;      # demanda anual
param K;      # costo por orden
param h;      # costo de mantener inventario por unidad por año
param S;      # costo por unidad en falta

var q >= 1, <= 10000;
var M >= 0, <= 10000;

minimize TotalCost:
    (K * D / q) + (h * M / 2) * (q / D) + (S * (q - M) / 2) * (q / D);

s.t. InventarioRealista: M <= q;

solve;

display q, M, TotalCost;

ampl: model problema10.mod;
data problema10.dat;
option solver ipopt;
solve;
display q, M, TotalCost;

EXIT: Optimal Solution Found.

Ipopt 3.12.13: Optimal Solution Found

suffix ipopt_zU_out OUT;
suffix ipopt_zL_out OUT;
ampl: q = 580.534
M = 580.534
TotalCost = 387.574
```

El modelo encontró que lo mejor es pedir unas 580 unidades por orden así, el costo total anual mínimo sería de aproximadamente \$387.57.

EJERCICIO 11

7 Every four years, Blockbuster Publishers revises its textbooks. It has been three years since the best-selling book, *The Joy of OR*, has been revised. At present, 2,000

copies of the book are in stock, and Blockbuster must determine how many copies of the book should be printed for the next year. The sales department believes that sales during the next year are governed by the distribution in Table 8. Each copy of *Joy* sold during the next year brings the publisher \$35 in revenues. Any copies left at the end of the next year cannot be sold at full price but can be sold for \$5 to Bonds Ennoble and Gitano's bookstores. The cost of a printing of the book is \$50,000 plus \$15 per book printed. How many copies of *Joy* should be printed? Would the answer change if 4,000 copies were currently in stock?

```
ampl: model problema11.mod;
ampl: data problema11.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 19900
0 simplex iterations
ampl: display UtilidadTotal;
UtilidadTotal = 19900
```

```
param p; # precio por unidad
param c; # costo por unidad
param F; # costo fijo
param d; # demanda máxima

var x >= 0, <= d; # cantidad a producir

maximize UtilidadTotal: p * x - c * x - F;
```

el modelo mostró que imprimir más es rentable y se logra una ganancia de \$19.900

EJERCICIO 12

8 Vivian and Wayne are planning on going to Lamaze natural childbirth classes. Lamaze classes meet once a week for five weeks. Each class gives 20% of the knowledge needed for “natural” childbirth. If Vivian and Wayne finish their classes before the birth of their child, they will forget during each week 5% of what they have learned in class. To maximize their expected knowledge at the time of childbirth, during which week of pregnancy should they begin classes? Assume that the number of weeks from conception to childbirth follows the probability distribution given in Table 9.

TABLE 9

Week of Birth	Probability
36	.05
37	.15
39	.20
40	.30
41	.15
42	.10
43	.05

```

set semanas ordered;

param Prob{semanas};
param clase_duracion;
param conocimiento_clase;
param alpha;

param semana_min := first(semanas);
param semana_max := last(semanas);

var inicio_clases integer >= semana_min, <= semana_max - clase_duracion + 1;

var clase_dada{s in semanas} binary;

subject to ClasesActivas {
    s in semanas
}:
    clase_dada[s] = if (ord(s) >= inicio_clases) and (ord(s) < inicio_clases + clase_duracion) then 1 else 0;

var conocimiento_ret{s in semanas} >= 0;

subject to CalculoConocimientoRet {
    s in semanas
}:
    conocimiento_ret[s] = sum {j in semanas}
        (if (ord(j) >= inicio_clases) and (ord(j) < inicio_clases + clase_duracion) and (ord(j) <= ord(s)) then
            conocimiento_clase * (1 - alpha)^(ord(s) - ord(j)) * clase_dada[j]
            else 0);

maximize ConocimientoEsperado:
    sum {s in semanas} Prob[s] * conocimiento_ret[s];

ampl: model problema12.mod;
ampl: data problema12.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:          Gurobi 12.0.1: optimal solution; objective -0
0 simplex iterations
ampl: display ConocimientoEsperado;
ConocimientoEsperado = 0

```

Vivian y Wayne quieren saber en qué semana del embarazo deberían comenzar sus clases de parto para recordar lo máximo posible al momento del nacimiento de su bebé. Sin embargo, el modelo muestra que, con los datos actuales, cualquier momento que elijan resulta en que olvidan todo antes del parto. En otras palabras, según el modelo, no vale la pena tomar las clases bajo esas condiciones. Esto sugiere que deberían revisar el calendario, el nivel de olvido o incluso considerar repetir las clases más cerca del parto.

EJERCICIO 13

9⁺ Some universities allow an employee to put an amount q into an account at the beginning of each year, to be used for child-care expenses. The amount q is not subject to federal income tax. Assume that all other income is taxed by the federal government at a 40% rate. If child-care expenses for the year (call them d) are less than q , the employee in effect loses $q - d$ dollars in before-tax income. If child-care expenses exceed q , the employee must pay the excess out of his or her own pocket but may credit 25% of that as a savings on his or her state income tax.

Suppose Professor Muffy Rabbit believes that there is an equal chance that her child-care expenses for the coming year will be \$3,000, \$4,000, \$5,000, \$6,000, or \$7,000. At the beginning of the year, how much money should she place in the child-care account?

```

set D;                                #niveles de gasto
param p{D};                           #probabilidades de cada nivel de gasto
param tasa_imp;                       #tasa de impuestos
param tasa_credito;                   #tasa de credito
param q_max;

param g{D};                           #gasto

var q >= 0, <= q_max;

var UtilidadEscenario{d in D} =
    if q >= g[d] then tasa_imp * (q - g[d])
    else - tasa_credito * (g[d] - q);

maximize UtilidadEsperada:
    sum{d in D} p[d] * UtilidadEscenario[d];

ampl: model problema13.mod;
ampl: data problema13.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                        Gurobi 12.0.1: optimal solution; objective 2000
0 simplex iterations
ampl: display UtilidadEsperada;
UtilidadEsperada = 2000

```

El resultado muestra que lo mejor es ahorrar \$5,000, ya que así logra el mayor beneficio financiero esperado (\$2,000), equilibrando el ahorro en impuestos con el riesgo de gastar más o menos de lo previsto.

EJERCICIO 14

5[†] Motorama TV estimates the annual demand for its TVs is (and will be in the future) normally distributed, with a mean of 6,000 and standard deviation of 2,000. Motorama

must determine how much production capacity it should have. The cost of building enough production capacity to make 1,000 sets per year is \$1,000,000 (equivalent in present value terms to a cost of \$100,000 per year forever). Exclusive of the cost of building capacity, each set sold contributes \$250 to profits. How much production capacity should Motorama have?

```
param mu;                #demanda esperada
param sigma;             #desviacion estandar de la demanda
param utilidad;          #utilidad por unidad vendida
param costo_capacidad;   #costo anual

set D;

param z {D};
param p {D};

var x >= 0;              #capacidad

param d_real {d in D} := mu + sigma * z[d];

maximize UtilidadEsperada:
    sum {d in D}
        p[d] * utilidad * ( if d_real[d] <= x * 1000 then d_real[d] else x * 1000 )
        - costo_capacidad * x;

ampl: model problema14.mod;
ampl: data problema14.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 553825
23 simplex iterations
1 branching node
ampl: display UtilidadEsperada;
UtilidadEsperada = 553825
```

El modelo recomienda que instalen capacidad para producir 8,000 unidades al año

EJERCICIO 15

6 I. L. Pea is a well-known mail-order company. During the Christmas rush (from November 1 to December 15), the number of orders that I. L. Pea must fill each day (five days per week) is normally distributed, with a mean of 2,000 and a standard deviation of 500. I. L. Pea must determine how many employees should be working during the Christmas rush. Each employee works five days a week, eight hours a day, can process 50 orders per day, and is paid \$10 per hour. If the full-time work force cannot handle the day's orders during regular hours, some employees will have to work overtime. Each employee is paid \$15 per hour for overtime work. For example, if 300 orders are received in a day and there are four employees, then $300 - 4(50) = 100$ orders must be processed by employees who are working overtime. Since each employee can fill $\frac{50}{8} = 6.25$ orders per hour, I. L. Pea would need to pay workers $\frac{100}{6.25} = 16$ hours of overtime for that day. To minimize its expected labor costs, how many full-time employees should I. L. Pea employ during the Christmas rush?

```
set Escenarios;

param mu;
param sigma;
param z {Escenarios};
param p {Escenarios};

param horas_dia = 8;
param dias_semana = 5;
param ordenes_por_dia = 50;
param pago_regular = 10;
param pago_extra = 15;

param horas_por_orden := 1 / ordenes_por_dia;

var x >= 0, integer; # cantidad de empleados a contratar

param demanda {i in Escenarios} := mu + sigma * z[i];

var horas_totales {i in Escenarios} >= 0;
var costo_diario {i in Escenarios} >= 0;

s.t. DefHorasTotales {i in Escenarios}:
    horas_totales[i] >= (demanda[i] - x * ordenes_por_dia) * horas_por_orden;

s.t. DefCostoDiario {i in Escenarios}:
    costo_diario[i] = (x * horas_dia * pago_regular) + (horas_totales[i] * pago_extra);

var CostoEsperado;

s.t. DefinicionCosto:
    CostoEsperado = sum{i in Escenarios} p[i] * costo_diario[i];

minimize Objetivo: CostoEsperado;

ampl: model problema15.mod;
ampl: data problema15.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1: Gurobi 12.0.1: optimal solution; objective 712.2
0 simplex iterations
ampl: display CostoEsperado;
CostoEsperado = 712.2
```

El modelo recomienda contratar 20 empleados a tiempo completo y da como resultado un costo esperado mínimo de \$712,2 por día en salarios.

EJERCICIO 16

6 In this problem, use marginal analysis to determine the optimal reorder point for the lost sales case.

a Show that the average inventory level for the lost sales case may be written as

$$\frac{1}{2}[(r - E(\mathbf{X}) + E(\mathbf{B}_r)) + (r - E(\mathbf{X}) + E(\mathbf{B}_r) + q)] \\ = r - E(\mathbf{X}) + E(\mathbf{B}_r) + \frac{q}{2}$$

b Although expected orders per year will no longer equal $\frac{E(\mathbf{D})}{q}$ (why?), we assume that the expected number of lost sales per year is relatively small. Thus, we may still assume that expected orders per year $= \frac{E(\mathbf{D})}{q}$. Now use marginal analysis to derive (15).

```
param r;                #punto de reorden
param q >= 0;           #cantidad de pedido
param E_X;              #demanda
param E_Br;             #ventas perdidas
param D;                #demanda anual
param c;                #costo unitario
param h;                #costo de mantenimiento
param p;                #costo por unidad perdida

var q_dec >= 1;

var I_prom = 0.5 * (r - E_X + E_Br + q_dec);

var CostTotal = (D / q_dec) * c + h * I_prom + p * E_Br;

minimize CostoEsperado: CostTotal;

ampl: model problema16.mod;
ampl: data problema16.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:          Gurobi 12.0.1: optimal solution; objective 37428.20323
12 simplex iterations
7 branching nodes
absmipgap=3.1802, relmipgap=8.4968e-05
ampl: display CostoEsperado;
CostoEsperado = 37428.2
```

Se determina que el valor óptimo de pedido resulta en un costo total esperado de \$37,428.20.

EJERCICIO 17

5 A fur dealer buys fur coats for \$100 each and sells them for \$200 each. He believes that the demand for coats is $N(100, 100)$. Any coat not sold can be sold to a discount house for \$100, but the fur dealer believes he must charge himself a cost of 10¢ per dollar invested in a fur coat that is sold at discount. How many coats should the dealer order? If the price at which the dealer sold his coats increased (assuming demand is unchanged), would he buy more or fewer coats?

```
set ESCENAS;

param mu;
param sigma;
param p;
param c;
param s;
param h;
param z {ESCENAS};    # valores de z de la normal estándar
param prob {ESCENAS};  # probabilidades

param demanda_real {i in ESCENAS} := mu + sigma * z[i];

var q >= 0;

maximize UtilidadEsperada:
    sum {i in ESCENAS}
        prob[i] * (
            if q <= demanda_real[i] then
                (p - c) * q
            else
                (p - c) * demanda_real[i] + (s - c - h * c) * (q - demanda_real[i])
        );

ampl: model problema17.mod;
ampl: data problema17.dat;
ampl: option solver cplex;
ampl: solve;
CPLEX 22.1.2:          CPLEX 22.1.2: optimal solution; objective 9896.04
33 simplex iterations
ampl: display UtilidadEsperada;
UtilidadEsperada = 9896.04
```

La solución muestra que la utilidad esperada máxima es \$9.896

EJERCICIO 18

3 James Beard bakes cheesecakes and Black Forest cakes. During any month, he can bake at most 65 cakes. The costs per cake and the demands for cakes, which must be met on time, are listed in Table 33. It costs 50¢ to hold a cheesecake, and 40¢ to hold a Black Forest cake, in inventory for a month. Formulate an LP to minimize the total cost of meeting the next three months' demands.

TABLE 33

Item	Month 1		Month 2		Month 3	
	Demand	Cost/Cake (\$)	Demand	Cost/Cake (\$)	Demand	Cost/Cake (\$)
Cheesecake	40	3.00	30	3.40	20	3.80
Black Forest	20	2.50	30	2.80	10	3.40

```

set T ordered; #meses
set I;         #tipos de pasteles

param D{i in I, t in T}; #demanda de pasteles
param C{i in I, t in T}; #costo por pastel
param H{i in I};        #costo de 1 unidad en inventario
param Cap;              #capacidad mensual maxima de produccion

var x{i in I, t in T} >= 0; #cantidad producida
var s{i in I, t in T} >= 0; #inventario final
var y{i in I, t in T} >= 0; #satisfacer demanda

minimize TotalCost:
    sum{i in I, t in T} (C[i,t] * x[i,t] + H[i] * s[i,t]);

subject to
    Demanda{i in I, t in T}:
        y[i,t] = D[i,t];

subject to
    Balance{i in I, t in T: ord(t) = 1}:
        x[i,t] - y[i,t] = s[i,t];

subject to
    BalanceT{i in I, t in T: ord(t) > 1}:
        x[i,t] + s[i,prev(t)] - y[i,t] = s[i,t];

subject to
    Capacidad{t in T}:
        sum{i in I} x[i,t] <= Cap;

ampl: model ejercicio18.mod;
ampl: data ejercicio18.dat;
ampl: option solver cplex;
ampl: solve;
CPLEX 22.1.2:          CPLEX 22.1.2: optimal solution; objective 464.5
5 simplex iterations
ampl: display TotalCost;
TotalCost = 464.5

```

La solución muestra que, se puede cubrir toda la demanda gastando solo \$464.5 en total.

EJERCICIO 19

6 Gandhi Clothing Company produces shirts and pants. Each shirt requires 2 sq yd of cloth, each pair of pants, 3. During the next two months, the following demands for shirts and pants must be met (on time): month 1—10 shirts, 15 pairs of pants; month 2—12 shirts, 14 pairs of pants. During each month, the following resources are available: month 1—90 sq yd of cloth; month 2—60 sq yd. (Cloth that is available during month 1 may, if unused during month 1, be used during month 2.)

During each month, it costs \$4 to make an article of clothing with regular-time labor and \$8 with overtime labor. During each month, a total of at most 25 articles of clothing may be produced with regular-time labor, and an unlimited number of articles of clothing may be produced with overtime labor. At the end of each month, a holding cost of \$3 per article of clothing is assessed. Formulate an LP that can be used to meet demands for the next two months (on time) at minimum cost. Assume that at the beginning of month 1, 1 shirt and 2 pairs of pants are available.

```
set T := 1..2; #meses
set A := { "shirt", "pants" }; #articulos

param demanda {A, T}; #demanda por articulo y mes
param costo {A}; #costo normal
param costo_ot {A}; #costo horas extra
param inventario_inicial {A};
param capacidad; #limite por mes
param cost_inv; #costo inventario
param tela_disp {T}; #tela disponible
param uso_tela {A}; #tela usada

var x {A, T} >= 0, integer; #produccion normal
var x_ot {A, T} >= 0, integer; #produccion horas extra
var s {A, T} >= 0, integer; # Inventario final

#inventario mes 1
s.t. BalanceInicial {a in A}:
    x[a,1] + x_ot[a,1] + inventario_inicial[a] - demanda[a,1] = s[a,1];

#inventario mes 2
s.t. BalanceMes2 {a in A}:
    x[a,2] + x_ot[a,2] + s[a,1] - demanda[a,2] = s[a,2];

#restriccion de capacidad de produccion normal
s.t. CapacidadNormal {t in T}:
    sum {a in A} x[a,t] <= capacidad;

#restriccion de uso de tela
s.t. RestriccionTela {t in T}:
    sum {a in A} uso_tela[a] * (x[a,t] + x_ot[a,t]) <= tela_disp[t];

#minimizar costo total
minimize CostoTotal:
    sum {a in A, t in T} (costo[a] * x[a,t] + costo_ot[a] * x_ot[a,t] + cost_inv * s[a,t]);
```

```

ampl: model ejercicio19.mod;
ampl: data ejercicio19.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 198
6 simplex iterations
1 branching node
ampl: display CostoTotal;
CostoTotal = 198

```

Gandhi Clothing debe planificar la producción de camisas y pantalones durante dos meses, considerando límites de tela, capacidad de trabajo y costos. El modelo optimiza esta planificación y determina que el costo mínimo para cumplir con la demanda es de 198

EJERCICIO 20

10 Payoff Insurance Company charges a customer according to his or her accident history. A customer who has had no accident during the last two years is charged a \$100 annual premium. Any customer who has had an accident during each of the last two years is charged a \$400 annual premium. A customer who has had an accident during only one of the last two years is charged an annual premium of \$300. A customer who has had an accident during the last year has a 10% chance of having an accident during the current year. If a customer has not had an accident during the last year, there is only a 3% chance that he or she will have an accident during the current year. During a given year, what is the average premium paid by a Payoff customer? (*Hint*: In case of difficulty, try a four-state Markov chain.)

```

set ESTADOS;
param prima{ESTADOS} >= 0;
param P{ESTADOS, ESTADOS} >= 0;

var pi{ESTADOS} >= 0;

s.t. Balance{k in ESTADOS}:
    pi[k] = sum{j in ESTADOS} pi[j] * P[j,k];

s.t. Normalizacion:
    sum{k in ESTADOS} pi[k] = 1;

minimize PrimaEsperada:
    sum{k in ESTADOS} pi[k] * prima[k];

```

```

ampl: model problema20.mod;
ampl: data problema20.dat;
ampl: option solver cplex;
ampl: solve;
CPLEX 22.1.2:                               CPLEX 22.1.2: optimal solution; objective 107.2590242
0 simplex iterations
ampl: display PrimaEsperada;
PrimaEsperada = 107.259

```

Un cliente promedio de Payoff Insurance paga aproximadamente \$107.26 por año en primas.

Ejercicio 21

10 Bectol, Inc. is building a dam. A total of 10 million cu ft of dirt is needed to construct the dam. A bulldozer is used to collect dirt for the dam. Then the dirt is moved via dumpers to the dam site. Only one bulldozer is available, and it rents for \$100 per hour. Bectol can rent, at \$40 per hour, as many dumpers as desired. Each dumper can hold 1,000 cu ft of dirt. It takes an average of 12 minutes for the bulldozer to load a dumper with dirt, and each dumper an average of five minutes to deliver the dirt to the dam and return to the bulldozer. Making appropriate assumptions about exponentiality, determine how Bectol can minimize the total expected cost of moving the dirt needed to build the dam. (*Hint: There is a machine repair problem somewhere!*)

```
set K;                                #dumppers

param Q = 10000;                      #cantidad total de viajes
param c_d = 40;                       #costo hora dumper
param c_b = 100;                      #costo hora bulldozer
param t_carga = 0.2;                  # tiempo de cargar de un dumper
param t_viaje = 0.0833;               # tiempo de viaje y regreso (en horas)

param mu := 1 / t_carga;

param ciclo_dumper := t_carga + t_viaje;

param lambda {k in K} := k / ciclo_dumper;

param Wq {k in K} := if lambda[k] < mu then (lambda[k] * t_carga^2) / (1 - (lambda[k] * t_carga)) else 1e6;

param T_ciclo {k in K} := t_viaje + t_carga + Wq[k];

param T_total {k in K} := Q * T_ciclo[k] / k;

param Costo_total {k in K} := T_total[k] * (c_b + k * c_d);

var x {k in K} binary;                # 1 si se elige k dumpers, 0 en otro caso

minimize Costo_Minimo:
    sum {k in K} Costo_total[k] * x[k];

s.t. One_Choice:
    sum {k in K} x[k] = 1;

data;

ampl: model ejercicio21.mod;
ampl: data ejercicio21.dat;
ampl: option solver gurobi;
ampl: solve;
Solution determined by presolve;
objective Costo_Minimo = 1068888.908.
```

El costo mínimo esperado para mover toda la tierra y construir la represa es de \$1.068.888,91

Ejercicio 22

13 You have invented a new product: the HAL DVD player. Each of 1,000 potential customers places a different value on this product. A consumer's valuation is equally likely to be any number between \$0 and \$1,000. It costs \$100 to produce the HAL player. During a year in which we set a price p for the product, all customers valuing the product at $\$p$ or more will purchase the product. Each year, we set a price for the product. What pricing strategy will maximize our expected profit over three years? What commonly observed phenomenon does this problem illustrate?

```
set T ordered;                #años

param N = 1000;               #total de consumidores
param V_max = 1000;           #valor maximo
param C = 100;                #costo por unidad

var p {t in T} >= C, <= V_max; #precio en cada año

var f {t in T};               #fraccion que compra en año t
var r {t in T};               #fraccion que queda al inicio del año t

s.t. remanencia_1: r[1] = 1;   #inicializacion del primer año

s.t. remanencia {t in T: ord(t) > 1}:
    r[t] = r[prev(t)] - f[prev(t)];

s.t. fraccion {t in T}:
    f[t] = r[t] * (V_max - p[t]) / V_max;

#ganancia total
var Profit;

s.t. Total_Profit:
    Profit = sum {t in T} N * f[t] * (p[t] - C);

maximize Max_Profit: Profit;

ampl: reset;
ampl: model ejercicio22.mod;
ampl: data ejercicio22.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 407034.2639
67231 simplex iterations
16472 branching nodes
absmipgap=14.0155, relmipgap=3.44332e-05
ampl: display Max_Profit;
Max_Profit = 407034
```

El resultado indica que, siguiendo una estrategia óptima de precios a lo largo de tres años, la empresa puede obtener una ganancia total esperada de \$407.034.

Ejercicio 23

15 General Motors has three auto divisions (1, 2, and 3). It also has an accounting division and a management consulting division. The question is: What fraction of the cost of the accounting and management consulting divisions should be allocated to each auto division? We assume that the entire cost of the accounting and management consulting departments must be allocated to the three auto divisions. During a given year, the work of the accounting division and management consulting division is allocated as shown in Table 4.

For example, accounting spends 10% of its time on problems generated by the accounting department, 20% of its time on work generated by division 3, and so forth. Each year, it costs \$63 million to run the accounting department and \$210 million to run the management consulting department. What fraction of these costs should be allocated to each auto division? Think of \$1 in costs incurred in accounting work. There is a .20 chance that this dollar should be allocated to each auto division, a .30 chance it should be allocated to consulting, and a .10 chance to accounting. If the dollar is allocated to an auto division, we know which division should be charged for that dollar. If the dollar is charged to consulting (for example), we repeat the process until the dollar is eventually charged to an auto division. Use knowledge of absorbing chains to figure out how to allocate the costs of running the accounting and management consulting departments among the three auto divisions.

```
set DIVISIONES;
set ABSORBENTES within DIVISIONES;
set TRANSITORIAS within DIVISIONES;

param costo {TRANSITORIAS};
param P {DIVISIONES, DIVISIONES} >= 0, <= 1; #matriz de transicion de probabilidades entre divisiones

var prob_absorber {i in TRANSITORIAS, j in ABSORBENTES};

s.t. Ecuacion_Absorcion {i in TRANSITORIAS, j in ABSORBENTES}:
    prob_absorber[i,j] = sum {k in DIVISIONES}
        P[i,k] * (if k in ABSORBENTES then (if k = j then 1 else 0) else prob_absorber[k,j]);

var costo_asignado {j in ABSORBENTES} >= 0;

s.t. Asignacion_Costos {j in ABSORBENTES}:
    costo_asignado[j] = sum {i in TRANSITORIAS} costo[i] * prob_absorber[i,j];

maximize Sin_Objetivo: 0;
```

```
ampl: reset;
model ejercicio23.mod;
data ejercicio23.dat;
solve;
display prob_absorber;
display costo_asignado;
ampl: CPLEX 22.1.2:                CPLEX 22.1.2: optimal solution; objective 0
0 simplex iterations
prob_absorber :=
Consultoria  Auto1    0.371795
Consultoria  Auto2    0.371795
Consultoria  Auto3    0.25641
Contabilidad Auto1    0.346154
Contabilidad Auto2    0.346154
Contabilidad Auto3    0.307692
;

costo_asignado [*] :=
Auto1  99.8846
Auto2  99.8846
Auto3  73.2308
;
```

El resultado indica que, al distribuir los costos de los departamentos de contabilidad y consultoría, las divisiones automotrices deben asumir los siguientes montos: Auto1: \$99,88 millones, Auto2: \$99,88 millones y Auto3: \$73,23 millones.

Ejercicio 24

3 During the next three months, Steelco faces the following demands for steel: 100 tons (month 1); 200 tons (month 2); 50 tons (month 3). During any month, a worker can produce up to 15 tons of steel. Each worker is paid \$5,000 per month. Workers can be hired or fired at a cost of \$3,000 per worker fired and \$4,000 per worker hired (it takes 0 time to hire a worker). The cost of holding a ton of steel in inventory for one month is \$100. Demand may be backlogged at a cost of \$70 per ton month. That is, if 1 ton of month 1 demand is met during month 3, then a backlogging cost of \$140 is incurred. At the beginning of month 1, Steelco has 8 workers. During any month, at most 2 workers can be hired. All demand must be met by the end of month 3. The raw material used to produce a ton of steel costs \$300. Formulate an LP to minimize Steelco's costs.

```
set T ordered;

param demanda {T};
param produccion_por_trabajador;
param sueldo;
param contratacion;
param despido;
param inventario_costo;
param backlog_costo;
param backlog_costo_mes1_en_mes3;
param trabajadores_inicio;
param contrataciones_max;
param costo_materia_prima;

var trabajadores {T} >= 0, integer;
var contratados {T} >= 0, integer;
var despedidos {T} >= 0, integer;
var produccion {T} >= 0;
var inventario {T} >= 0;
var backlog {T} >= 0;

minimize CostoTotal:
    sum{t in T} (
        trabajadores[t] * sueldo
        + contratados[t] * contratacion
        + despedidos[t] * despido
        + inventario[t] * inventario_costo
        + (if t = 3 then backlog[t] * backlog_costo_mes1_en_mes3 else backlog[t] * backlog_costo)
        + produccion[t] * costo_materia_prima
    );

subject to ProduccionTotal {t in T}:
    produccion[t] = trabajadores[t] * produccion_por_trabajador;

subject to BalanceTrabajadores {t in T: ord(t) = 1}:
    trabajadores[t] = trabajadores_inicio + contratados[t] - despedidos[t];

subject to BalanceTrabajadoresT {t in T: ord(t) > 1}:
    trabajadores[t] = trabajadores[prev(t)] + contratados[t] - despedidos[t];

subject to MaxContratados {t in T}:
    contratados[t] <= contrataciones_max;

subject to BalanceMaterial {t in T: ord(t) = 1}:
    produccion[t] + inventario[t] = demanda[t] + backlog[t];

subject to BalanceMaterialT {t in T: ord(t) > 1}:
    produccion[t] + inventario[prev(t)] + backlog[prev(t)] = demanda[t] + inventario[t] + backlog[t];
```

```

ampl: model ejercicio24.mod;
ampl: data ejercicio24.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:          Gurobi 12.0.1: optimal solution; objective 58250
3 simplex iterations
1 branching node
ampl: display CostoTotal;
CostoTotal = 58250

```

El resultado indica que el costo total mínimo para que la empresa Steelco satisfaga la demanda de acero durante los tres meses es de \$58.250

Ejercicio 25

58† To process income tax forms, the IRS first sends each form through the data preparation (DP) department, where information is coded for computer entry. Then the form is sent to data entry (DE), where it is entered into the computer. During the next three weeks, the following number of forms will arrive: week 1, 40,000; week 2, 30,000; week 3, 60,000. The IRS meets the crunch by hiring employees who work 40 hours per week and are paid \$200 per week. Data preparation of a form requires 15 minutes, and data entry of a form requires 10 minutes. Each week, an employee is assigned to either data entry or data preparation. The IRS must complete processing of all forms by the end of week 5 and wants to minimize the cost of accomplishing this goal. Formulate an LP that will determine how many workers should be working each week and how the workers should be assigned over the next five weeks.

```

set S := 1..5;          #semanas

param llegada {s in S} >= 0;

param horas_semana = 40;
param costo_trabajador = 200;

param t_dp = 0.25;      #tiempo de preparacion
param t_de = 1/6;       #tiempo de entrada

var tr_dp {s in S} >= 0, integer;      #trabajadores DP en semana s
var tr_de {s in S} >= 0, integer;      #trabajadores DE en semana s

var preparados {s in S} >= 0;
var ingresados {s in S} >= 0;

var inventario {s in S} >= 0;

s.t. capacidad_dp {s in S}:
    preparados[s] <= tr_dp[s] * horas_semana / t_dp;

s.t. capacidad_de {s in S}:
    ingresados[s] <= tr_de[s] * horas_semana / t_de;

s.t. inventario_semana_1:
    inventario[1] = preparados[1] - ingresados[1];

s.t. inventario_siguiete {s in 2..5}:
    inventario[s] = inventario[s-1] + preparados[s] - ingresados[s];

s.t. total_preparado:
    sum {s in S} preparados[s] = sum {s in S} llegada[s];

s.t. total_ingresado:
    sum {s in S} ingresados[s] = sum {s in S} llegada[s];

minimize CostoTotal:
    sum {s in S} costo_trabajador * (tr_dp[s] + tr_de[s]);

```

```

ampl: model ejercicio25.mod;
ampl: data ejercicio25.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:          Gurobi 12.0.1: optimal solution; objective 271000
12 simplex iterations
1 branching node
ampl: display CostoTotal;
CostoTotal = 271000

```

El resultado indica que el costo mínimo total para procesar todos los formularios de impuestos durante las cinco semanas es de \$271.000

Ejercicio 26

7 Each hour from 10 A.M. to 7 P.M., Bank One receives checks and must process them. Its goal is to process all the checks the same day they are received. The bank has 13 check-processing machines, each of which can process up to 500 checks per hour. It takes one worker to operate each machine. Bank One hires both full-time and part-time workers. Full-time workers work 10 A.M.–6 P.M., 11 A.M.–7 P.M., or Noon–8 P.M. and are paid \$160 per day. Part-time workers work either 2 P.M.–7 P.M. or 3 P.M.–8 P.M. and are paid \$75 per day. The number of checks received each hour is given in Table 6. In the interest of maintaining continuity, Bank One believes it must have at least three full-time workers under contract. Develop a cost-minimizing work schedule that processes all checks by 8 P.M.

TABLE 6

Time	Checks Received
10 A.M.	5,000
11 A.M.	4,000
Noon	3,000
1 P.M.	4,000
2 P.M.	2,500
3 P.M.	3,000
4 P.M.	4,000
5 P.M.	4,500
6 P.M.	3,500
7 P.M.	3,000

```

var x >= 3, integer;    #full-time workers
var y1 >= 0, integer;   #part-time (2pm-7pm)
var y2 >= 0, integer;   #part-time (3pm-8pm)

set H := 10..19;        #horas

param req {H};          #cheques recibidos

s.t. h10: x              >= req[10];
s.t. h11: x              >= req[11];
s.t. h12: x              >= req[12];
s.t. h13: x              >= req[13];
s.t. h14: x + y1         >= req[14];
s.t. h15: x + y1 + y2    >= req[15];
s.t. h16: x + y1 + y2    >= req[16];
s.t. h17: x + y1 + y2    >= req[17];
s.t. h18: x              >= req[18];
s.t. h19: y2             >= req[19];

|
minimize CostoTotal:
    160 * x + 75 * y1 + 75 * y2;

```



```

ampl: model ejercicio26.mod;
ampl: data ejercicio26.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 2050
0 simplex iterations
ampl: display CostoTotal;
CostoTotal = 2050

```

El costo mínimo total para contratar trabajadores y procesar todos los cheques a tiempo en Bank One es de \$2.050.

Ejercicio 27

6 Daisy Drugs manufactures two drugs: 1 and 2. The drugs are produced by blending two chemicals: 1 and 2. By weight, drug 1 must contain at least 65% chemical 1, and drug 2 must contain at least 55% chemical 1. Drug 1 sells for \$6/oz, and drug 2 sells for \$4/oz. Chemicals 1 and 2 can be produced by one of two production processes. Running process 1 for an hour requires 3 oz of raw material and 2 hours skilled labor and yields 3 oz of each chemical. Running process 2 for an hour requires 2 oz of raw material and 3 hours of skilled labor and yields 3 oz of chemical 1 and 1 oz of chemical 2. A total of 120 hours of skilled labor and 100 oz of raw material are available. Formulate an LP that can be used to maximize Daisy's sales revenues.

```

var x1 >= 0;    #horas de proceso 1
var x2 >= 0;    #horas de proceso 2

var d1 >= 0;    #onzas de droga 1
var d2 >= 0;    #onzas de droga 2

var c1;        #cantidad químico 1
var c2;        #cantidad químico 2

s.t. produccion_q1:
    c1 = 3 * x1 + 3 * x2;

s.t. produccion_q2:
    c2 = 3 * x1 + 1 * x2;

s.t. uso_q1:
    0.65 * d1 + 0.55 * d2 <= c1;

s.t. uso_q2:
    0.35 * d1 + 0.45 * d2 <= c2;

s.t. trabajo:
    2 * x1 + 3 * x2 <= 120;    #horas de trabajo

s.t. materia:
    3 * x1 + 2 * x2 <= 100;    #onzas de materia prima

maximize ingreso:
    6 * d1 + 4 * d2;

```

```

ampl: model ejercicio27.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 1200
3 simplex iterations
ampl: display ingreso;
ingreso = 1200

```

El ingreso máximo total que Daisy Drugs puede obtener al producir y vender sus dos drogas, dadas las restricciones de recursos, es de \$1.200.

Ejercicio 28

7¹ Lizzie's Dairy produces cream cheese and cottage cheese. Milk and cream are blended to produce these two products. Both high-fat and low-fat milk can be used to produce cream cheese and cottage cheese. High-fat milk is 60% fat; low-fat milk is 30% fat. The milk used to produce cream cheese must average at least 50% fat and that for cottage cheese, at least 35% fat. At least 40% (by weight) of the inputs to cream cheese and at least 20% (by weight) of the inputs to cottage cheese must be cream. Both cottage cheese and cream cheese are produced by putting milk and cream through the cheese machine. It costs 40¢ to process 1 lb of inputs into a pound of cream cheese. It costs 40¢ to produce 1 lb of cottage cheese, but every pound of input for cottage cheese yields 0.9 lb of cottage cheese and 0.1 lb of waste. Cream can be produced by evaporating high-fat and low-fat milk. It costs 40¢ to evaporate 1 lb of high-fat milk. Each pound of high-fat milk that is evaporated yields 0.6 lb of cream. It costs 40¢ to evaporate 1 lb of low-fat milk. Each pound of low-fat milk that is evaporated yields 0.3 lb of cream. Each day, up to 3,000 lb of input may be sent through the cheese machine. Each day, at least 1,000 lb of cottage cheese and 1,000 lb of cream cheese must be produced. Up to 1,500 lb of cream cheese and 2,000 lb of cottage cheese can be sold each day. Cottage cheese is sold for \$1.20/lb and cream cheese for \$1.50/lb. High-fat milk is purchased for 80¢/lb and low-fat milk for 40¢/lb. The evaporator can process at most 2,000 lb of milk daily. Formulate an LP that can be used to maximize Lizzie's daily profit.

```

var leche_alta >= 0;           #leche alta en grasa
var leche_baja >= 0;           #leche baja en grasa

var evap_alta >= 0;            #leche alta en grasa enviada al evaporador
var evap_baja >= 0;            #leche baja en grasa enviada al evaporador

var queso_crema >= 0;          #libras de queso crema producido
var queso_cottage >= 1000;     #libras de cottage producido

var crema_a_crema >= 0;        #crema usada en queso crema
var crema_a_cottage >= 0;      #crema usada en cottage

var leche_a_crema >= 0;        #leche usada en queso crema
var leche_a_cottage >= 0;      #leche usada en cottage

var crema_total = 0.6 * evap_alta + 0.3 * evap_baja; #crema producida
var leche_total = leche_alta + leche_baja;

param grasa_alta := 0.60;
param grasa_baja := 0.30;
param grasa_crema := 1.00;

```

```

s.t. grasa_queso_crema:
    (grasa_alta * leche_a_crema + grasa_baja * leche_a_crema + grasa_crema * crema_a_crema)
    / (leche_a_crema + crema_a_crema) >= 0.5;

s.t. grasa_cottage:
    (grasa_alta * leche_a_cottage + grasa_baja * leche_a_cottage + grasa_crema * crema_a_cottage)
    / (leche_a_cottage + crema_a_cottage) >= 0.35;

s.t. crema_en_crema:
    crema_a_crema / (leche_a_crema + crema_a_crema) >= 0.4;

s.t. crema_en_cottage:
    crema_a_cottage / (leche_a_cottage + crema_a_cottage) >= 0.2;

s.t. balance_leche:
    leche_a_crema + leche_a_cottage + evap_alta + evap_baja <= leche_alta + leche_baja;

s.t. uso_crema:
    crema_a_crema + crema_a_cottage <= crema_total;

s.t. capacidad_maquina:
    leche_a_crema + crema_a_crema + leche_a_cottage + crema_a_cottage <= 3000;

s.t. rendimiento_crema:
    queso_crema = leche_a_crema + crema_a_crema;

s.t. rendimiento_cottage:
    queso_cottage = 0.9 * (leche_a_cottage + crema_a_cottage);

s.t. max_crema_crema:
    crema_a_crema <= 1500;

s.t. max_crema_cottage:
    crema_a_cottage <= 1000;

s.t. demanda_crema:
    queso_crema <= 2000;

s.t. limite_evaporador:
    evap_alta + evap_baja <= 2000;

maximize ganancia:
    1.50 * queso_crema + 1.20 * queso_cottage
    - 0.80 * leche_alta - 0.40 * leche_baja
    - 0.40 * (evap_alta + evap_baja);

ampl: model ejercicio28.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1: Gurobi 12.0.1: optimal solution; objective 1920.740741
5 simplex iterations
1 branching node
ampl: display ganancia;
ganancia = 1920.74

```

la ganancia diaria máxima que Lizzie's Dairy puede obtener al producir queso crema y queso cottage, respetando las restricciones de materia prima, contenido graso, evaporación y demanda, es de \$1.920

Ejercicio 29

9 Each week Chemco can purchase unlimited quantities of raw material at \$6/lb. Each pound of purchased raw material can be used to produce either input 1 or input 2. Each pound of raw material can yield 2 oz of input 1, requiring 2 hours of processing time and incurring \$2 in processing costs. Each pound of raw material can yield 3 oz of input 2, requiring 2 hours of processing time and incurring \$4 in processing costs.

Two production processes are available. It takes 2 hours to run process 1, requiring 2 oz of input 1 and 1 oz of input 2. It costs \$1 to run process 1. Each time process 1 is run 1 oz of product A and 1 oz of liquid waste are produced. Each time process 2 is run requires 3 hours of processing time, 2 oz of input 2 and 1 oz of input 1. Process 2 yields 1 oz of product B and .8 oz of liquid waste. Process 2 incurs \$8 in costs.

Chemco can dispose of liquid waste in the Port Charles River or use the waste to produce product C or product D. Government regulations limit the amount of waste Chemco is allowed to dump into the river to 1,000 oz/week. One ounce of product C costs \$4 to produce and sells for \$11. One hour of processing time, 2 oz of input 1, and .8 oz of liquid waste are needed to produce an ounce of product C. One unit of product D costs \$5 to produce and sells for \$7. One hour of processing time, 2 oz of input 2, and 1.2 oz of liquid waste are needed to produce an ounce of product D.

At most 5,000 oz of product A and 5,000 oz of product B can be sold each week, but weekly demand for products C and D is unlimited. Product A sells for \$18/oz and product B sells for \$24/oz. Each week 6,000 hours of processing time is available. Formulate an LP whose solution will tell Chemco how to maximize weekly profit.

```
var r >= 0;           #materia prima comprada
var x1 >= 0;          #input 1
var x2 >= 0;          #input 2
var pa >= 0;          #veces que se ejecuta A
var pb >= 0;          #veces que se ejecuta B
var a >= 0;           #producto A producido
var b >= 0;           #producto B producido
var waste >= 0;       #residuos generados
var pc >= 0;          #producto C producido
var pd >= 0;          #producto D producido

param costo_mat := 0.6;
param venta_a := 18;
param venta_b := 24;
param venta_c := 11;
param venta_d := 7;
param costo_c := 4;
param costo_d := 5;

var tiempo_total = 2 * pa + 3 * pb + pc + pd;

s.t. materia_total:
    x1 + x2 <= r;

s.t. input1: x1 = 2 * r / (1 + 2);
s.t. input2: x2 = 2 * r / (2 + 2);

s.t. produccion_a:
    a = pa;

s.t. produccion_b:
    b = pb;

s.t. residuos:
    waste = pa + 0.8 * pb;

s.t. uso_residuos:
    0.8 * pc + 0.8 * pd <= waste;

s.t. limite_residuo:
    0.8 * pc + 0.8 * pd <= 1000;

s.t. input_c:
    2 * pc <= x1;

s.t. input_d:
    2 * pd <= x2;

s.t. limite_tiempo:
    tiempo_total <= 6000;

s.t. limite_a: a <= 5000;
s.t. limite_b: b <= 5000;

maximize ganancia:
    venta_a * a + venta_b * b + venta_c * pc + venta_d * pd
    - costo_mat * r - costo_c * pc - costo_d * pd;

ampl: model ejercicio29.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 54000
0 simplex iterations
ampl: display ganancia;
ganancia = 54000
```

La ganancia semanal máxima que Chemco puede obtener, considerando las restricciones de materia prima, tiempo de procesamiento, límites de producción y disposición de residuos, es de \$54.000.

Ejercicio 30

9 Donovan Enterprises produces electric mixers. During the next four quarters, the following demands for mixers must be met on time: quarter 1—4,000; quarter 2—2,000; quarter 3—3,000; quarter 4—10,000. Each of Donovan's workers works three quarters of the year and gets one quarter off. Thus, a worker may work during quarters 1, 2, and 4 and get quarter 3 off. Each worker is paid \$30,000 per year and (if working) can produce up to 500 mixers during a quarter. At the end of each quarter, Donovan incurs a holding cost of \$30 per mixer on each mixer in inventory. Formulate an LP to help Donovan minimize the cost (labor and inventory) of meeting the next year's demand (on time). At the beginning of quarter 1, 600 mixers are available.

```
set Q := 1..4;

param d {Q};           # demanda
param inventario_inicial := 600;
param capacidad := 500;
param salario := 30000;
param costo_inv := 30;

#
var w1 >= 0, integer;
var w2 >= 0, integer;
var w3 >= 0, integer;
var w4 >= 0, integer;

var inv {Q} >= 0;

var prod {Q} >= 0;

s.t. prod_q1: prod[1] = capacidad * (w1 + w2 + w3);
s.t. prod_q2: prod[2] = capacidad * (w1 + w2 + w4);
s.t. prod_q3: prod[3] = capacidad * (w1 + w3 + w4);
s.t. prod_q4: prod[4] = capacidad * (w2 + w3 + w4);

s.t. balance_1:
    prod[1] + inventario_inicial - d[1] = inv[1];

s.t. balance_rest {q in 2..4}:
    prod[q] + inv[q-1] - d[q] = inv[q];

minimize CostoTotal:
    salario * (w1 + w2 + w3 + w4)
    + sum {q in Q} costo_inv * inv[q];

ampl: model ejercicio30.mod;
ampl: data ejercicio30.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:           Gurobi 12.0.1: optimal solution; objective 597000
5 simplex iterations
1 branching node
ampl: display CostoTotal;
CostoTotal = 597000
```

El costo total mínimo para que Donovan Enterprises satisfaga la demanda de mixers durante los cuatro trimestres es de \$597.000.

Ejercicio 31

7 Suppose we are borrowing \$1,000 at 12% annual interest with 60 monthly payments. Assume equal payments are made at the end of month 1, month 2, . . . month 60. We know that entering into Excel the function

$$= \text{PMT}(.01, 60, 1,000)$$

would yield the monthly payment (\$22.24).

It is instructive to use LP to determine the monthly payment. Let p be the (unknown) monthly payment. Each month we owe $.01 \cdot$ (our current unpaid balance) in interest. The remainder of our monthly payment is used to reduce the unpaid balance. For example, suppose we paid \$30 each month. At the beginning of month 1, our unpaid balance is \$1,000. Of our month 1 payment, \$10 goes to interest and \$20 to paying off the unpaid balance. Then we would begin month 2 with an unpaid balance of \$980. The trick is to use LP to determine the monthly payment that will pay off the loan at the end of month 60.

```
set T := 1..60;                # Meses

param interes := 0.01;         # 1% mensual
param prestamo := 1000;        # monto inicial del prestamo

var p >= 0;                    # pago mensual
var b {T} >= 0;                # saldo no pagado al inicio del mes t

s.t. saldo_inicial:
    b[1] = prestamo;

s.t. saldo_dinamico {t in 2..60}:
    b[t] = b[t-1] + interes * b[t-1] - p;

s.t. saldo_final:
    b[60] + interes * b[60] - p = 0;

minimize pago_minimo:
    p;
```

```
ampl: model ejercicio31.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 22.24444768
0 simplex iterations
ampl: display pago_minimo;
pago_minimo = 22.2444
```

El pago mensual mínimo necesario para amortizar completamente el préstamo es de aproximadamente \$22,24.

Ejercicio 32

52[†] A paper-recycling plant processes box board, tissue paper, newsprint, and book paper into pulp that can be used to produce three grades of recycled paper (grades 1, 2, and 3). The prices per ton and the pulp contents of the four inputs are shown in Table 82. Two methods, de-inking and asphalt dispersion, can be used to process the four inputs into pulp. It costs \$20 to de-ink a ton of any input. The process of de-inking removes 10% of the input's pulp, leaving 90% of the original pulp. It costs \$15 to apply asphalt dispersion to a ton of material. The asphalt dispersion process removes 20% of the input's pulp. At most, 3,000 tons of input can be run through the asphalt dispersion process or the de-inking process. Grade 1 paper can only be produced with newsprint or book paper pulp; grade 2 paper, only with book paper, tissue paper, or box board pulp; and grade 3 paper, only with newsprint, tissue paper, or box board pulp. To meet its current demands, the company needs 500 tons of pulp for grade 1 paper, 500 tons of pulp for grade 2 paper, and 600 tons of pulp for grade 3 paper. Formulate an LP to minimize the cost of meeting the demands for pulp.

```

set I;      #tipos de papel
set P;      #procesos
set G;      #grados

param costo {P};
param rendimiento {P};
param demanda {G};

param permitido {I, G} binary;

var x {I, P, G} >= 0;

s.t. capacidad_total:
    sum {i in I, p in P, g in G} x[i,p,g] <= 3000;

s.t. satisfacer_demanda {g in G}:
    sum {i in I, p in P: permitido[i,g]=1} rendimiento[p] * x[i,p,g] >= demanda[g];

s.t. no_permitido {i in I, g in G, p in P}:
    if permitido[i,g] = 0 then x[i,p,g] = 0;

minimize costo_total:
    sum {i in I, p in P, g in G} costo[p] * x[i,p,g];

```

```

ampl: model ejercicio32.mod;
ampl: data ejercicio32.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                  Gurobi 12.0.1: optimal solution; objective 30000
3 simplex iterations
ampl: display costo_total;
costo_total = 30000

```

El costo mínimo total para satisfacer la demanda de los tres grados de papel reciclado, utilizando los distintos tipos de papel y procesos permitidos, es de \$30.000

Ejercicio 33

53 Turkeyco produces two types of turkey cutlets for sale to fast-food restaurants. Each type of cutlet consists of white meat and dark meat. Cutlet 1 sells for \$4/lb and must consist of at least 70% white meat. Cutlet 2 sells for \$3/lb and must consist of at least 60% white meat. At most, 50 lb of cutlet 1 and 30 lb of cutlet 2 can be sold. The two types of turkey used to manufacture the cutlets are purchased from the GobbleGobble Turkey Farm. Each type 1 turkey costs \$10 and yields 5 lb of white meat and 2 lb of dark meat. Each type 2 turkey costs \$8 and yields 3 lb of white meat and 3 lb of dark meat. Formulate an LP to maximize Turkeyco's profit.

```
var x1 >= 0;           #pavos tipo 1
var x2 >= 0;           #pavos tipo 2

var c1 >= 0;           #cutlet 1 producido
var c2 >= 0;           #cutlet 2 producido

var w1 >= 0;           #carne blanca en cutlet 1
var d1 >= 0;           #carne oscura en cutlet 1

var w2 >= 0;           #carne blanca en cutlet 2
var d2 >= 0;           #carne oscura en cutlet 2

s.t. definicion_c1: c1 = w1 + d1;
s.t. definicion_c2: c2 = w2 + d2;

s.t. blanco_min_c1: w1 >= 0.7 * c1;
s.t. blanco_min_c2: w2 >= 0.6 * c2;

s.t. limite_c1: c1 <= 50;
s.t. limite_c2: c2 <= 30;

s.t. blanca_usada: w1 + w2 <= 5*x1 + 3*x2;
s.t. oscura_usada: d1 + d2 <= 2*x1 + 3*x2;

maximize ganancia:
    4 * c1 + 3 * c2 - (10 * x1 + 8 * x2);
```

```
ampl: model ejercicio33.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 177.5555556
2 simplex iterations
ampl: display ganancia;
ganancia = 177.556
```

La ganancia máxima que Turkeyco puede obtener al producir y vender cortes de pavo es de \$177

Ejercicio 34

22 Monroe County is trying to determine where to place the county fire station. The locations of the county's four major towns are given in Figure 31. Town 1 is at (10, 20); town 2 is at (60, 20); town 3 is at (40, 30); town 4 is at (80, 60). Town 1 averages 20 fires per year; town 2, 30 fires; town 3, 40 fires; and town 4, 25 fires. The county wants to build the fire station in a location that minimizes the average distance that a fire engine must travel to respond to a fire. Since most roads run in either an east-west or a north-south direction, we assume that the fire engine can only do the same. Thus, if the fire station were located at (30, 40) and a fire occurred at town 4, the fire engine would have to travel $(80 - 30) + (60 - 40) = 70$ miles to the fire. Use linear programming to determine where the fire station should be located. (*Hint:* If the fire station is to be located at the point (x, y) and there is a town at the point (a, b) , define variables e, w, n, s (east, west, north, south) that satisfy the equations $x - a = w - e$ and $y - b = n - s$. It should now be easy to obtain the correct LP formulation.)

```

set I := 1..4;

param a {I};      # coordenada x de la ciudad i
param b {I};      # coordenada y de la ciudad i
param p {I};

var x;
var y;

var e {I} >= 0;    #este
var o {I} >= 0;    #oeste
var n {I} >= 0;    #norte
var s {I} >= 0;    #sur

s.t. distancia_x {i in I}:
    x - a[i] = e[i] - o[i];

s.t. distancia_y {i in I}:
    y - b[i] = n[i] - s[i];

minimize distancia_total:
    sum {i in I} p[i] * (e[i] + o[i] + n[i] + s[i]);

ampl: model ejercicio34.mod;
ampl: data ejercicio34.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                    Gurobi 12.0.1: optimal solution; objective 3450
5 simplex iterations
ampl: display distancia_total;
distancia_total = 3450

```

La distancia total mínima ponderada que un camión de bomberos recorrería para atender incendios es de 3.450 millas por año.

Ejercicio 35

25 Ghostbusters, Inc., exorcises (gets rid of) ghosts. During each of the next three months, the company will receive the following number of calls from people who want their ghosts exorcised: January, 100 calls; February, 300 calls; March, 200 calls. Ghostbusters is paid \$800 for each ghost exorcised during the month in which the customer calls. Calls need not be responded to during the month they are made, but if a call is responded to one month after it is made, then Ghostbusters loses \$100 in future goodwill, and if a call is responded to two months after it is made, Ghostbusters loses \$200 in goodwill. Each employee of Ghostbusters can exorcise 10 ghosts during a month. Each employee is paid a salary of \$4,000 per month. At the beginning of January, the company has 8 workers. Workers can be hired and trained (in 0 time) at a cost of \$5,000 per worker. Workers can be fired at a cost of \$4,000 per worker. Formulate an LP to maximize Ghostbusters' profit (revenue less costs) over the next three months. Assume that all calls must be handled by the end of March.

```
set T := 1..3;           #meses

param demanda {T};      #llamadas recibidas por mes
param capacidad := 10;   #llamadas que atiende un trabajador por mes
param salario := 4000;
param contratar := 5000;
param despedir := 4000;

param ingreso {i in T, j in T} :=
  if j = i then 800
  else if j = i + 1 then 700
  else if j = i + 2 then 600
  else 0;

var w {T} >= 0, integer;    #trabajadores disponibles
var h {T} >= 0, integer;    #contrataciones
var f {T} >= 0, integer;    #despidos

var x {i in T, j in T: j >= i && j <= i+2} >= 0;

s.t. satisfacer_llamados {i in T}:
  sum {j in T: j >= i && j <= i+2} x[i,j] = demanda[i];

s.t. capacidad_trabajo {t in T}:
  sum {i in T: i <= t && t <= i+2} x[i,t] <= capacidad * w[t];

s.t. trabajadores_evolucion {t in T: t > 1}:
  w[t] = w[t-1] + h[t] - f[t];

s.t. trabajadores_inicial:
  w[1] = 8;

maximize ganancia:
  sum {i in T, j in T: j >= i && j <= i+2} ingreso[i,j] * x[i,j]
  - sum {t in T} (salario * w[t] + contratar * h[t] + despedir * f[t]);

ampl: model ejercicio35.mod;
ampl: data ejercicio35.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                  Gurobi 12.0.1: optimal solution; objective 142000
4 simplex iterations
1 branching node
ampl: display ganancia;
ganancia = 142000
```

La ganancia máxima que Ghostbusters Inc. puede obtener durante los próximos tres meses, es de \$142.000.

Ejercicio 36

32 Alcoa produces 100-, 200-, and 300-foot-long aluminum ingots for customers. This week's demand for ingots is shown in Table 115.

Alcoa has 4 furnaces in which ingots can be produced. During a week, each furnace can be operated for 50 hours. Because ingots are produced by cutting long strips of aluminum, longer ingots take less time to produce than shorter ingots. If a furnace is devoted completely to producing one type of ingot; the number it can produce in a week is shown in Table 116.

For example, furnace 1 could produce 350 300-foot ingots per week. The material in an ingot costs \$10 per foot. If a customer wants a 100- or 200-foot ingot, then she will accept an ingot of that length or longer. How can Alcoa minimize the material costs incurred in meeting required weekly demands?

TABLE 116

Furnace	Ingot Length		
	100'	200'	300'
1	230	340	350
2	230	260	280
3	240	300	310
4	200	280	300

```

set H := 1..4;           #hornos
set L := {100, 200, 300}; #tipos de lingotes

param demanda {L};      #demanda
param produccion_max {H, L}; # capacidad de produccion

#numero de lingotes producidos
var x {H, L} >= 0, integer;

#si el horno produce ese tipo
var y {H, L} binary;

s.t. exclusivo {h in H}:
    sum {l in L} y[h,l] <= 1;

s.t. capacidad {h in H, l in L}:
    x[h,l] <= produccion_max[h,l] * y[h,l];

s.t. demanda_100:
    sum {h in H, l in L: l >= 100} x[h,l] >= demanda[100];

s.t. demanda_200:
    sum {h in H, l in L: l >= 200} x[h,l] >= demanda[200];

s.t. demanda_300:
    sum {h in H, l in L: l = 300} x[h,l] >= demanda[300];

minimize costo_total:
    sum {h in H, l in L} x[h,l] * l * 10;
.

ampl: model ejercicio36.mod;
ampl: data ejercicio36.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 900000
11 simplex iterations
1 branching node
ampl: display costo_total;
costo_total = 9e+05

```

el costo mínimo total de materiales necesario para satisfacer la demanda semanal de lingotes de aluminio es de \$900.000

Ejercicio 37

9 In the Bloomington Girls Club basketball league, the following games have been played: team A beat team B by 7 points, team C beat team A by 8 points, team B beat team C by 6 points, and team B beat team C by 9 points. Let A , B , and C represent “ratings” for each team in the sense that if, say, team A plays team B, then we predict that team A will defeat team B by $A - B$ points. Determine values of A , B , and C that best fit (in the least squares sense) these results. To obtain a unique set of ratings, it may be helpful to add the constraint $A + B + C = 0$. This ensures that an “average” team will have a rating of 0.

```

var A;
var B;
var C;

var e1;
var e2;
var e3;
var e4;

s.t. juego1: A - B = 7 + e1;
s.t. juego2: C - A = 8 + e2;
s.t. juego3: B - C = 6 + e3;
s.t. juego4: B - C = 9 + e4;

s.t. promedio: A + B + C = 0;

minimize error_total: e1^2 + e2^2 + e3^2 + e4^2;

ampl: model ejercicio37.mod;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 207
0 simplex iterations
2 barrier iterations
ampl: display error_total;
error_total = 207

```

El error cuadrático total mínimo al ajustar los ratings de los equipos A, B y C para que expliquen los resultados de los partidos jugados es de 207.

Ejercicio 38

4 A business believes that its needs for cash during the next month are described by the random variable shown in Table 23. At the beginning of the month, the business has \$10,000 available, and the business manager must determine how much of the money should be placed in an account bearing 24% annual interest. If any money must be withdrawn before the end of the month, all interest on the withdrawn money is forfeited, and a penalty equal to 2% of the withdrawn money must be paid. How much money should be placed in the 24% annual interest account?

TABLE 23

Cash Needs	Probability
\$4,000	.30
\$5,000	.20
\$6,000	.10
\$7,000	.30
\$8,000	.10

```

set E;                                #escenarios posibles

param nec {E};                        #necesidad
param prob {E};                       #probabilidad

param disponible := 10000;
param interes := 0.02;                #2% mensual
param penalizacion := 0.02;

var x >= 0;                            #dinero invertido
var w {e in E} >= 0;                  #dinero retirado anticipadamente

s.t. retiro_necesario {e in E}:
    w[e] >= nec[e] - (disponible - x);

s.t. retiro_maximo {e in E}:
    w[e] <= x;

maximize utilidad_esperada:
    sum {e in E}
        prob[e] * (
            disponible - nec[e]
            - penalizacion * w[e]
            + (x - w[e]) * interes
        );

ampl: model ejercicio38.mod;
ampl: data ejercicio38.dat;
ampl: option solver gurobi;
ampl: solve;
Gurobi 12.0.1:                    Gurobi 12.0.1: optimal solution; objective 4360
5 simplex iterations
ampl: display utilidad_esperada;
utilidad_esperada = 4360

```

La utilidad esperada máxima que puede obtener el negocio es de \$4.360

Ejercicio 39

11 A bank is trying to determine which of two machines should be rented to process checks. Machine 1 rents for \$10,000 per year and processes 1,000 checks per hour. Machine 2 rents for \$15,000 per year and processes 1,600 checks per hour. Assume that the machines work 8 hours a day, 5 days a week, 50 weeks a year. The bank must process an average of 800 checks per hour, and the average check processed is for \$100. Assume an annual interest rate of 20%. Then determine the cost to the bank (in lost interest) for each hour that a check spends waiting for and undergoing processing. Assuming that interarrival times and service times are exponential, which machine should the bank rent?

```
set M;  
  
param lambda := 800;           #tasa de llegada  
param valor_cheque := 100;  
param interes_anual := 0.20;  
param horas := 2000;  
  
param mu{M};                   #tasa de servicio  
param arriendo{M};             #costo de arriendo anual  
  
param costo_interes := valor_cheque * interes_anual; #costo por hora en sistema  
  
param tiempo_total{m in M} := 1 / (mu[m] - lambda);  
  
param costo_total{m in M} := arriendo[m] + lambda * tiempo_total[m] * costo_interes * horas;  
  
var elegir{m in M} binary;  
  
s.t. una_maquina: sum{m in M} elegir[m] = 1;  
  
minimize costo_total_banco: sum{m in M} costo_total[m] * elegir[m];  
  
ampl: model ejercicio39.mod;  
ampl: data ejercicio39.dat;  
ampl: option solver gurobi;  
ampl: solve;  
Gurobi 12.0.1:                Gurobi 12.0.1: optimal solution; objective 55000  
0 simplex iterations  
ampl: display costo_total_banco;  
costo_total_banco = 55000
```

El costo total mínimo anual para el banco, considerando el arriendo de la máquina y el costo del tiempo que los cheques pasan en el sistema, es de \$55.000.

Ejercicio 40

10 Coalco produces coal at three mines and ships it to four customers. The cost per ton of producing coal, the ash and sulfur content (per ton) of the coal, and the production capacity (in tons) for each mine are given in Table 16. The number of tons of coal demanded by each customer are given in Table 17.

The cost (in dollars) of shipping a ton of coal from a mine to each customer is given in Table 18. It is required that the total amount of coal shipped contain at most 5% ash and at most 4% sulfur. Formulate an LP that minimizes the cost of meeting customer demands.

TABLE 16

Mine	Production Cost (\$)	Capacity	Ash Content (Tons)	Sulfur Content (Tons)
1	50	120	.08	.05
2	55	100	.06	.04
3	62	140	.04	.03

TABLE 17

Customer 1	Customer 2	Customer 3	Customer 4
80	70	60	40

TABLE 18

Mine	Customer			
	1	2	3	4
1	4	6	8	12
2	9	6	7	11
3	8	12	3	5

```

set MINAS;
set CLIENTES;

param cap{MINAS};           #capacidad por mina
param c_prod{MINAS};        #costo producción por mina
param ceniza{MINAS};         #contenido de ceniza por mina
param azufre{MINAS};        #contenido de azufre por mina
param demanda{CLIENTES};    #demanda
param c_envio{MINAS, CLIENTES}; #costo envío

var x{MINAS, CLIENTES} >= 0; # cantidad enviada desde mina i a cliente j

minimize CostoTotal:
    sum{i in MINAS, j in CLIENTES} (c_prod[i] + c_envio[i,j]) * x[i,j];

subject to CapacidadMina{i in MINAS}:
    sum{j in CLIENTES} x[i,j] <= cap[i];

subject to DemandaCliente{j in CLIENTES}:
    sum{i in MINAS} x[i,j] = demanda[j];

subject to CenizaTotal:
    sum{i in MINAS, j in CLIENTES} ceniza[i] * x[i,j] <= 0.05 * sum{i in MINAS, j in CLIENTES} x[i,j];

subject to AzufreTotal:
    sum{i in MINAS, j in CLIENTES} azufre[i] * x[i,j] <= 0.04 * sum{i in MINAS, j in CLIENTES} x[i,j];

ampl: model ejercicio40.mod;
ampl: data ejercicio40.dat;
ampl: option model gurobi;
ampl: solve;
Gurobi 12.0.1:                               Gurobi 12.0.1: optimal solution; objective 16060
6 simplex iterations
ampl: display CostoTotal;
CostoTotal = 16060

```

El costo mínimo total para que Coalco satisfaga la demanda de carbón de los cuatro clientes, es de \$16.060